



Customer segmentation strategy based on purchase data

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Content

Introduction

Customer Data Collection & Data

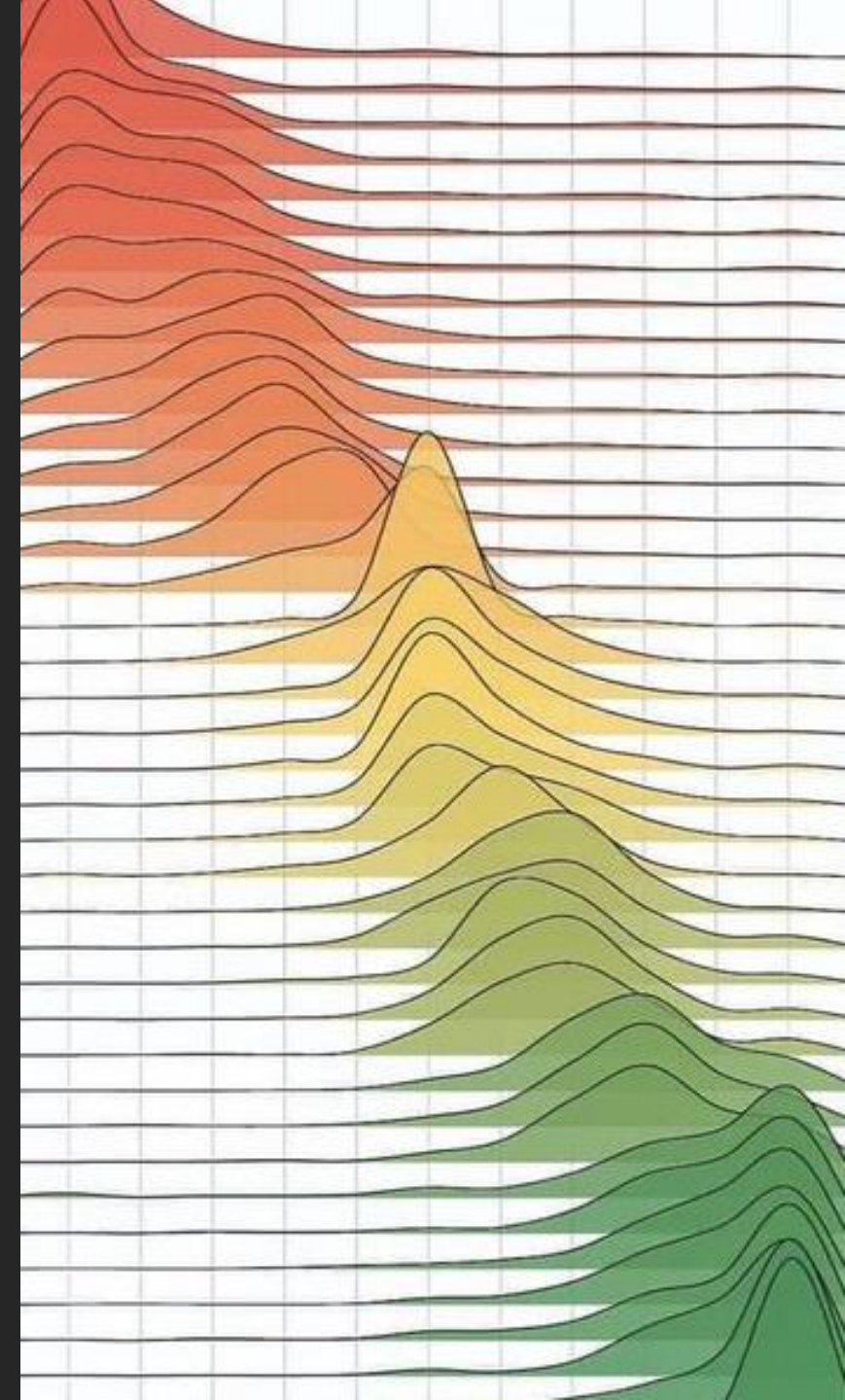
Cleaning

Data Visualization

Results

Segmentation

Marketing Strategy



INTRODUCTION

- **Objective:**

To leverage customer purchase data for creating targeted marketing strategies that increase engagement, loyalty, and revenue.

- **Focus:**

Using statistical tools to get insights about purchase pattern and subsequent development of marketing strategy based on segmentation.

CUSTOMER DATA COLLECTION

Amazon shopping
data is collected in
the format specified



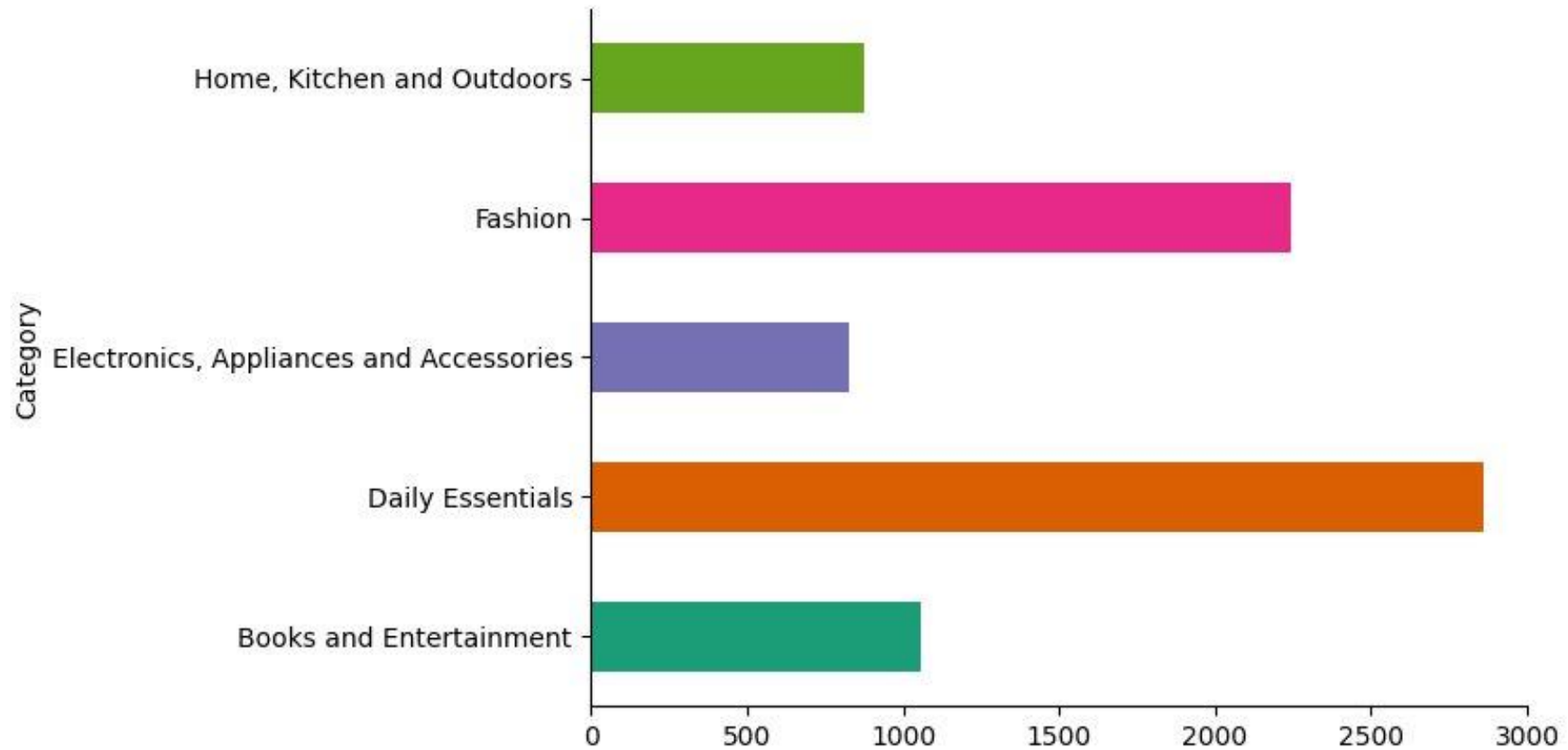
Initially we had 7865
data points collected
across the class



Data cleaning and
outlier removal
resulted in 7665 data
points

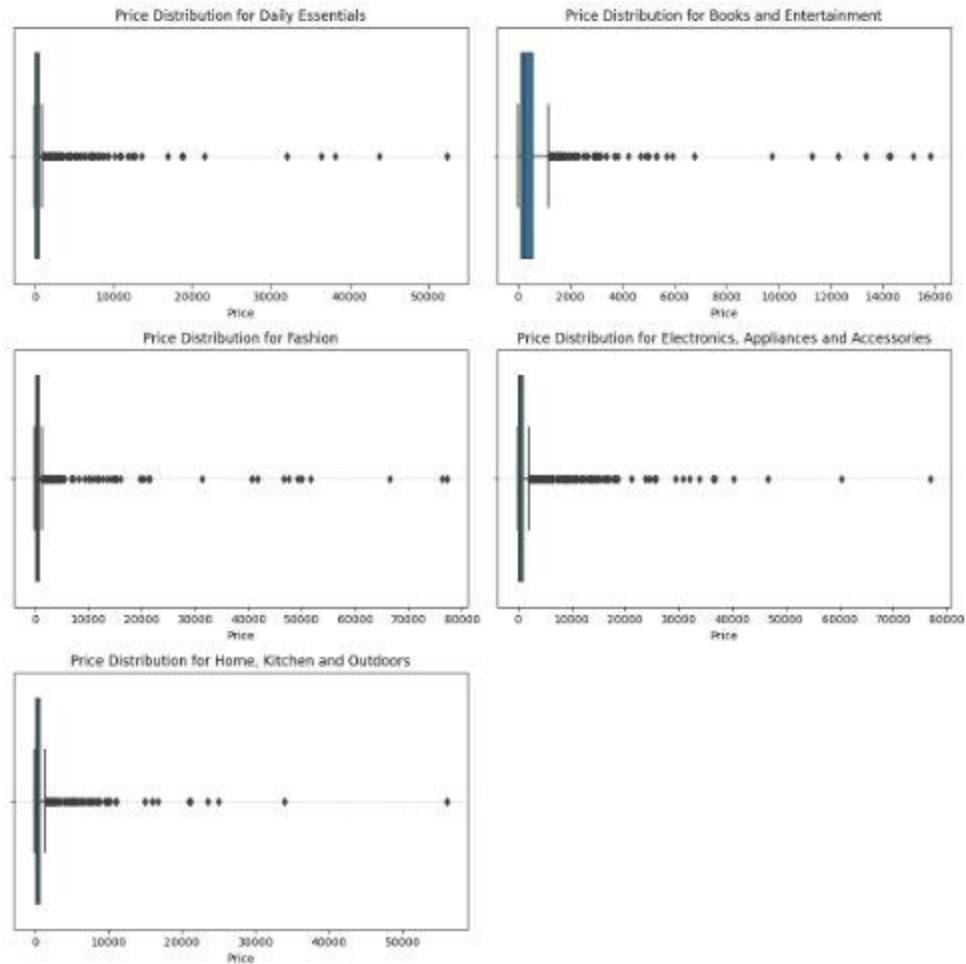
DATA VISUALIZATION

CATEGORY WISE NUMBER OF PURCHASES

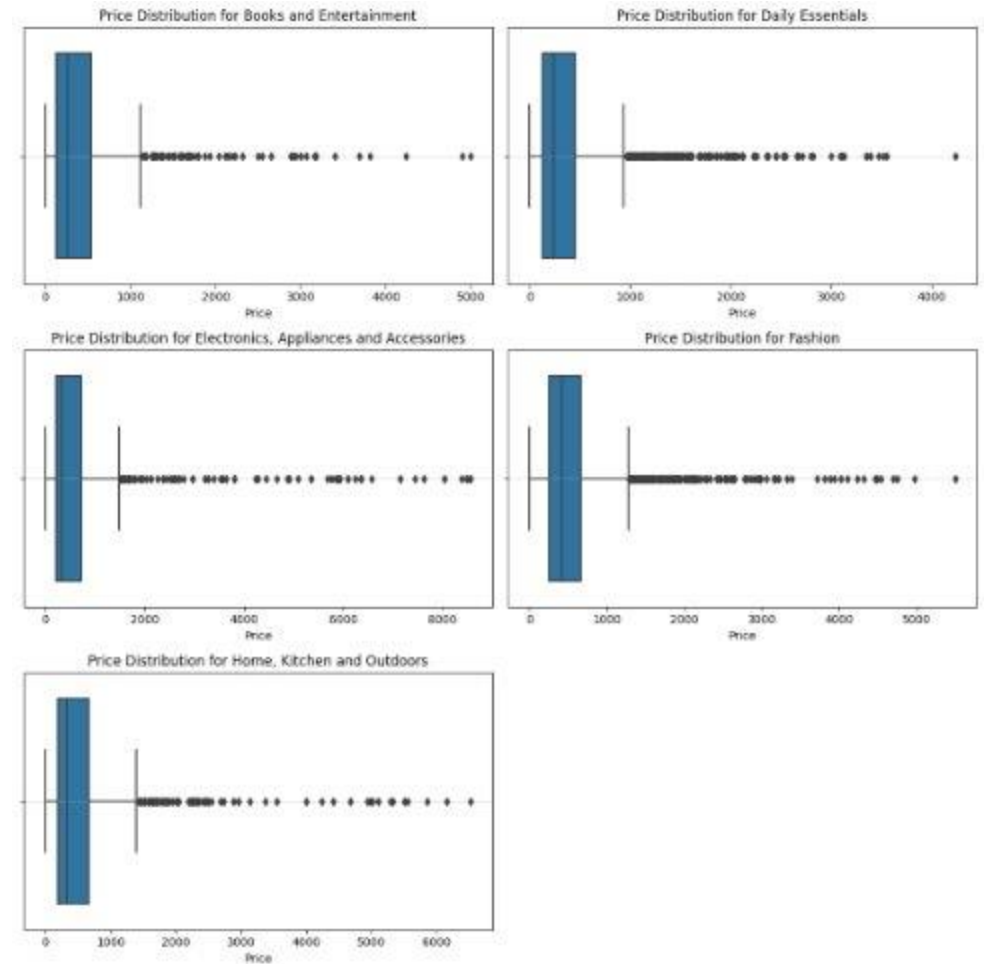


CATEGORY WISE PRICE DISTRIBUTION

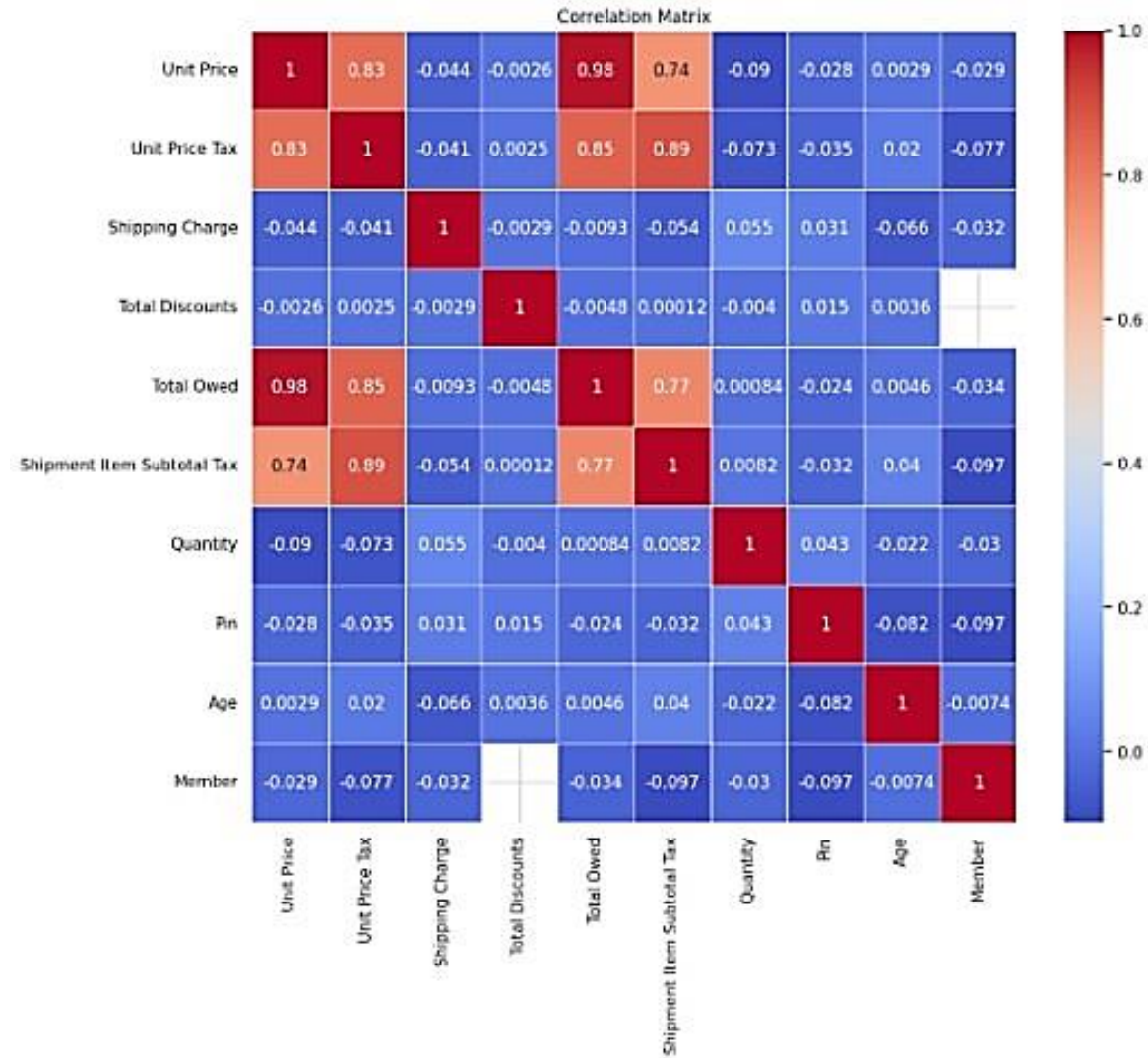
Before Outlier removal



After Outlier removal:



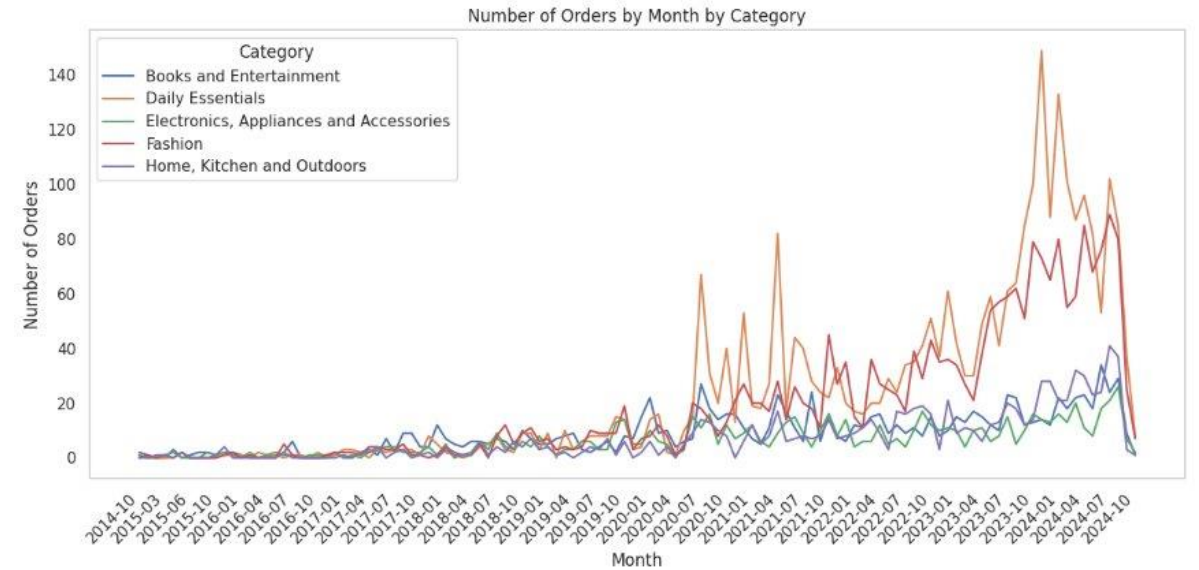
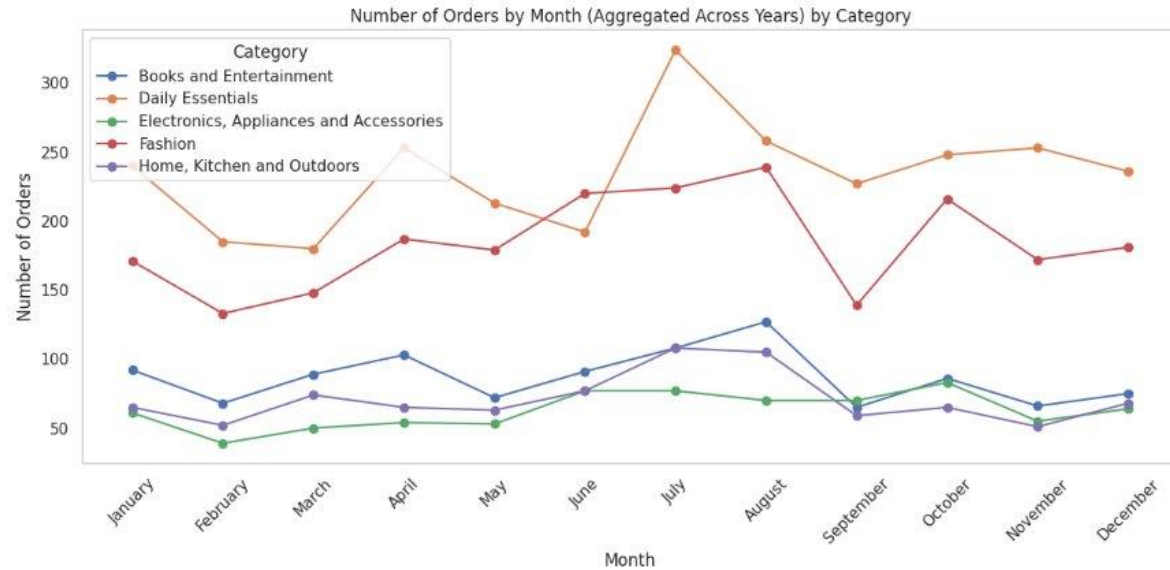
HEAT MAP OF CORRELATION MATRIX



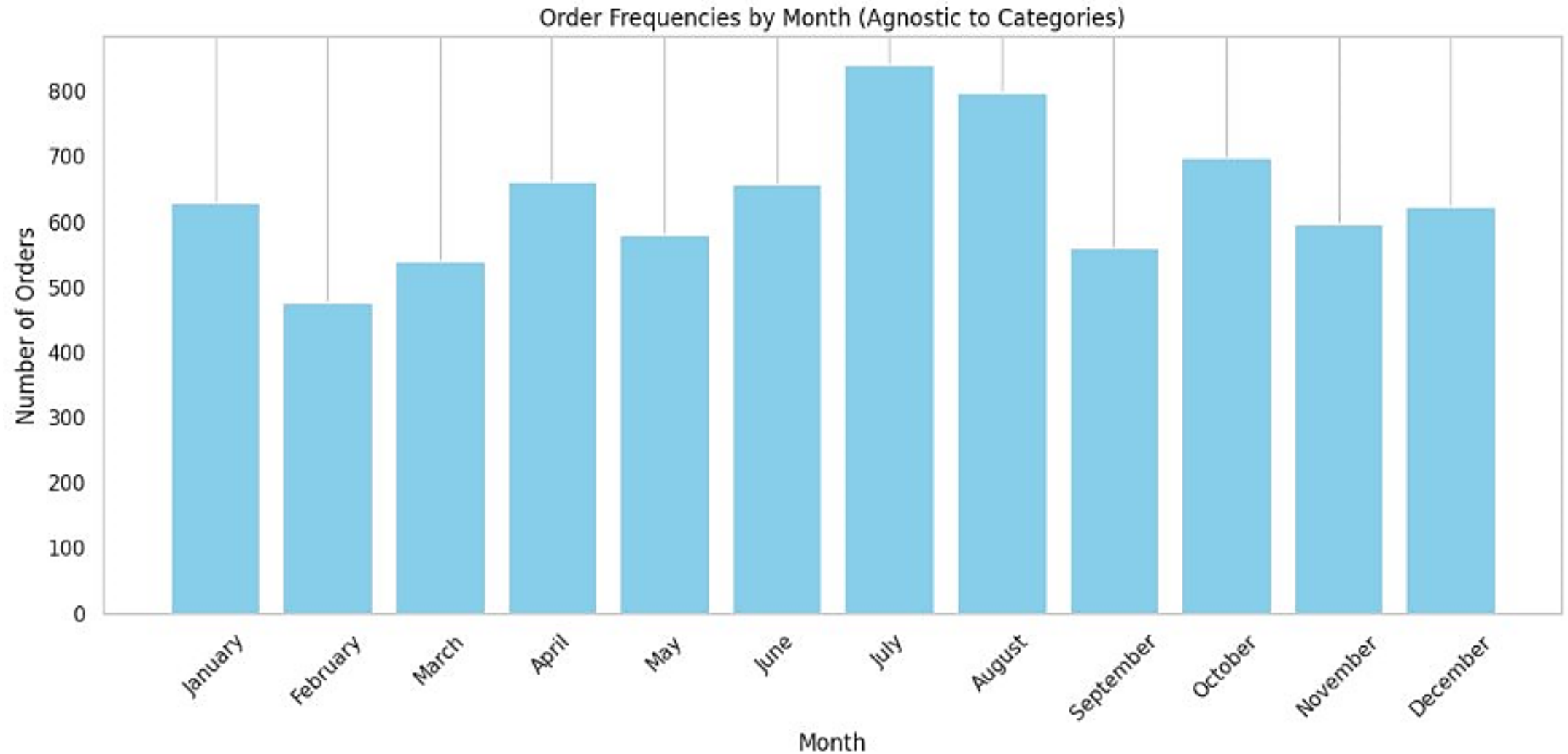
HISTOGRAM OF CATEGORY WISE PRICE



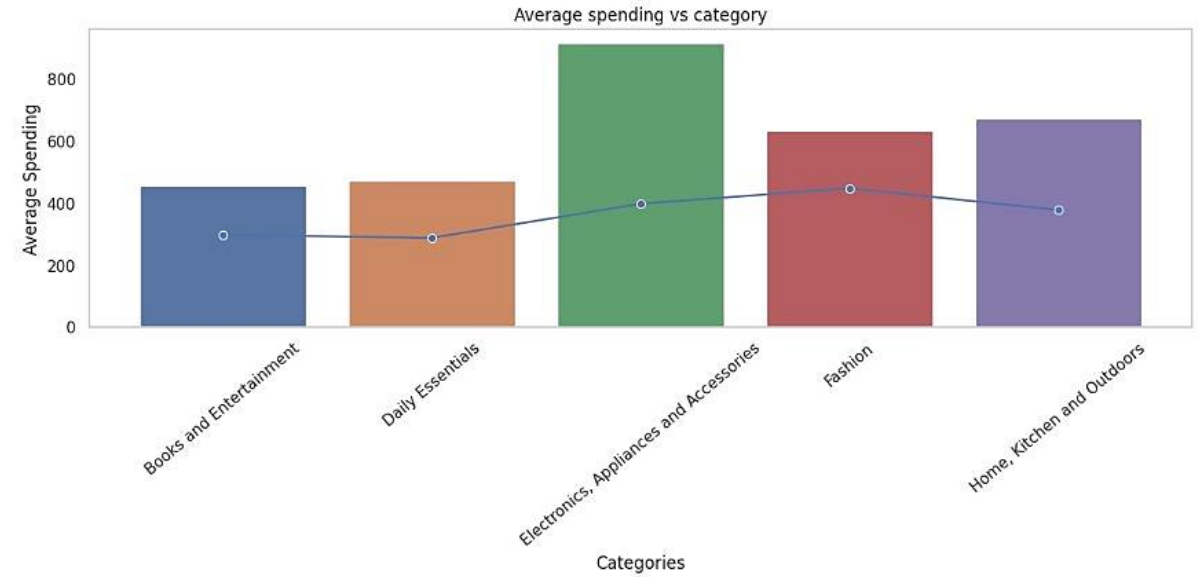
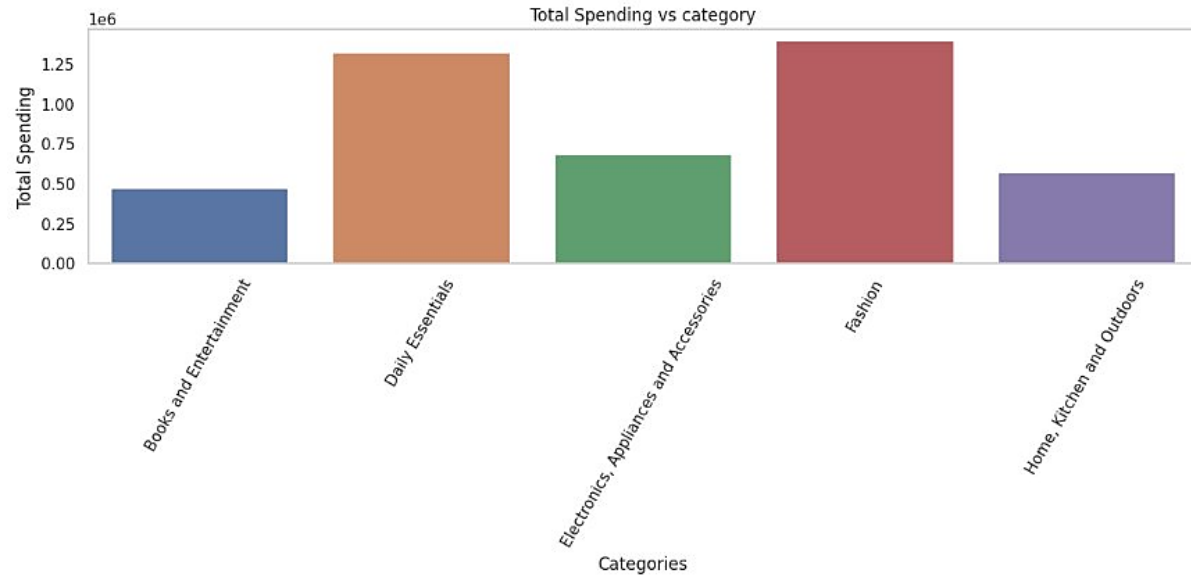
TIME SERIES OF ORDERS CATEGORY WISE



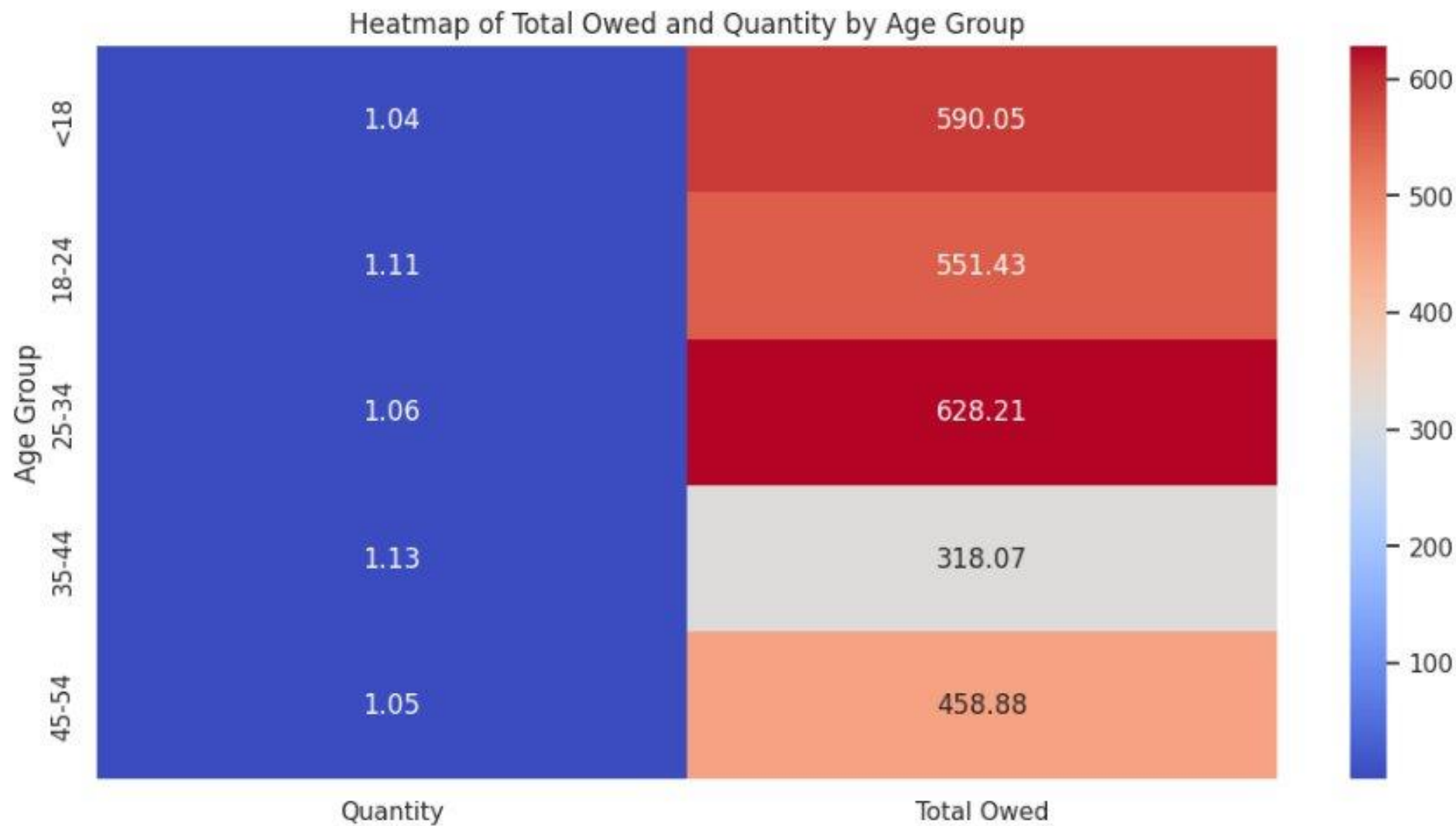
TOTAL ORDER BY MONTH



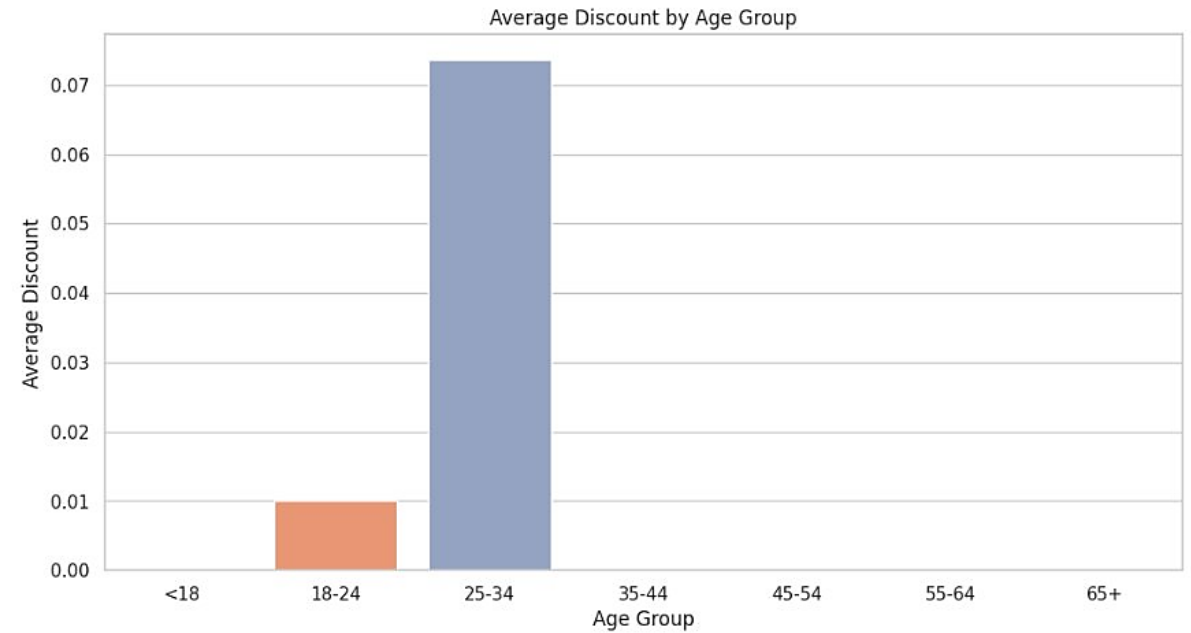
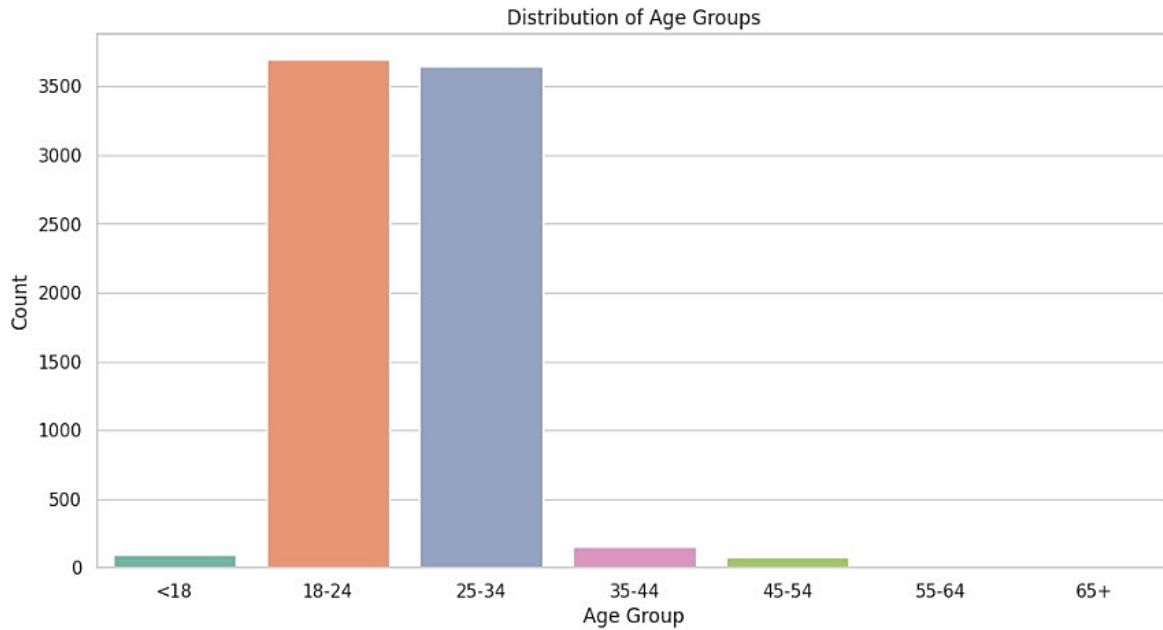
AVERAGE AND TOTAL SPEND VS CATEGORY



HEAT MAP OF QTY. & TOTAL SPEND WITH AGE



DISTRIBUTION OF ORDERS ACROSS AGE GROUP



STATISTICAL TOOLS

1. Welch's t-test : Compares the means of two independent samples, especially when variances are unequal.

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

- $\bar{X}_1, \bar{X}_2$: sample means of groups 1 and 2
- s_1^2, s_2^2 : sample variances
- n_1, n_2 : sample sizes

2. Chi-Square test : Tests the association between categorical variables or assesses if observed data fits an expected distribution.

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

O : Observed frequency in each category
 E : Expected frequency in each category

RESULT OF TEST

```
from scipy.stats import ttest_ind

# Filter data by gender
male_orders = df[df['Gender'] == 'M']['Total Owed']
female_orders = df[df['Gender'] == 'F']['Total Owed']

# Welch's t-test for Total Owed between genders
t_stat, p_val = ttest_ind(male_orders, female_orders, equal_var=False)
print(f"Welch's t-test: t-statistic = {t_stat}, p-value = {p_val}")
```

Welch's t-test: t-statistic = 6.519460505479551, p-value = 7.551319131340045e-11

```
import pandas as pd
from scipy.stats import chi2_contingency

# Create a crosstab for Mode of Payment and Gender
payment_gender_crosstab = pd.crosstab(df['Quantity'], df['Gender'])

# Chi-square test of independence
chi2, p, dof, ex = chi2_contingency(payment_gender_crosstab)
print(f"Chi-square test: chi2 = {chi2}, p-value = {p}")
```

Chi-square test: chi2 = 34.21384992991178, p-value = 3.715698986171419e-05

STATISTICAL TOOLS

3. Pearson's Correlation Coefficient : Measures the strength and direction of the linear relationship between two continuous variables.

Range: Values between -1 and +1.

- **+1:** Perfect positive linear relationship
- **-1:** Perfect negative linear relationship
- **0:** No linear relationship

$$r = \frac{\sum(X - \bar{X})(Y - \bar{Y})}{\sqrt{\sum(X - \bar{X})^2 \sum(Y - \bar{Y})^2}}$$

4. Power of the t-test : Indicates the probability of correctly rejecting the null hypothesis when it is false (i.e., detecting an actual effect).

$$\text{Power} = 1 - \beta$$

β : Probability of making a Type II error (failing to reject a false null hypothesis)

RESULT OF TEST

```
from scipy.stats import pearsonr
```

```
# Calculate Pearson's correlation for Age and Total Owed
corr, p_val = pearsonr(df['Age'], df['Total Owed'])
print(f"Pearson's correlation: correlation = {corr}, p-value = {p_val}")
```

Pearson's correlation: correlation = 0.03472220122249928, p-value = 0.00207159213011493

```
from statsmodels.stats.power import TTestIndPower
```

```
# Assuming medium effect size of 0.5, alpha level of 0.05, and sample size based on gender
power_analysis = TTestIndPower()
effect_size = 0.5
alpha = 0.05
sample_size = len(male_orders) # or a similar count

# Calculate power
power = power_analysis.solve_power(effect_size=effect_size, nobs1=sample_size, alpha=alpha, alternative='two-sided')
print(f"Power of the t-test: {power}")
```

Power of the t-test: 1.0

```
[32]: from scipy.stats import chi2_contingency
```

```
gender_cluster_crosstab = pd.crosstab(customer_data['Cluster'], df['Gender'])
chi2, p_value, _, _ = chi2_contingency(gender_cluster_crosstab)
print(f"Chi-square result: chi2 = {chi2}, p-value = {p_value}")
```

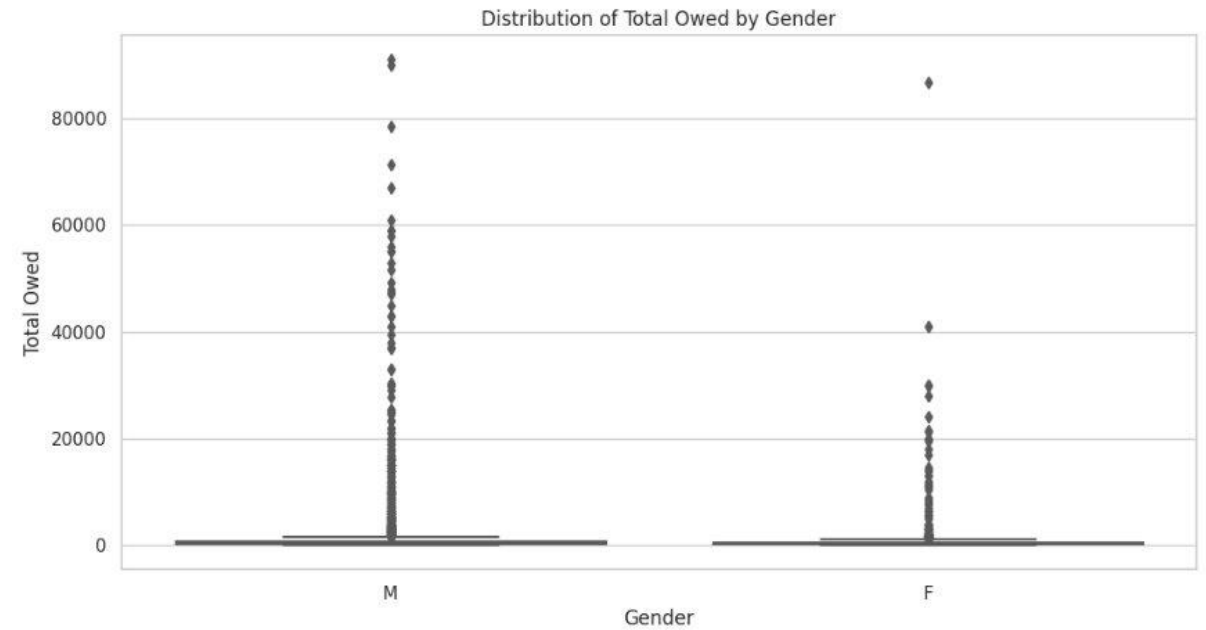
Chi-square result: chi2 = 0.0, p-value = 1.0

```
[33]: # Assuming 'Unit Price' can be used to classify the product
df['Price Category'] = pd.cut(df['Unit Price'], bins=[0, 50, 150, float('inf')], labels=['Low-priced', 'Economy', 'Premium'])

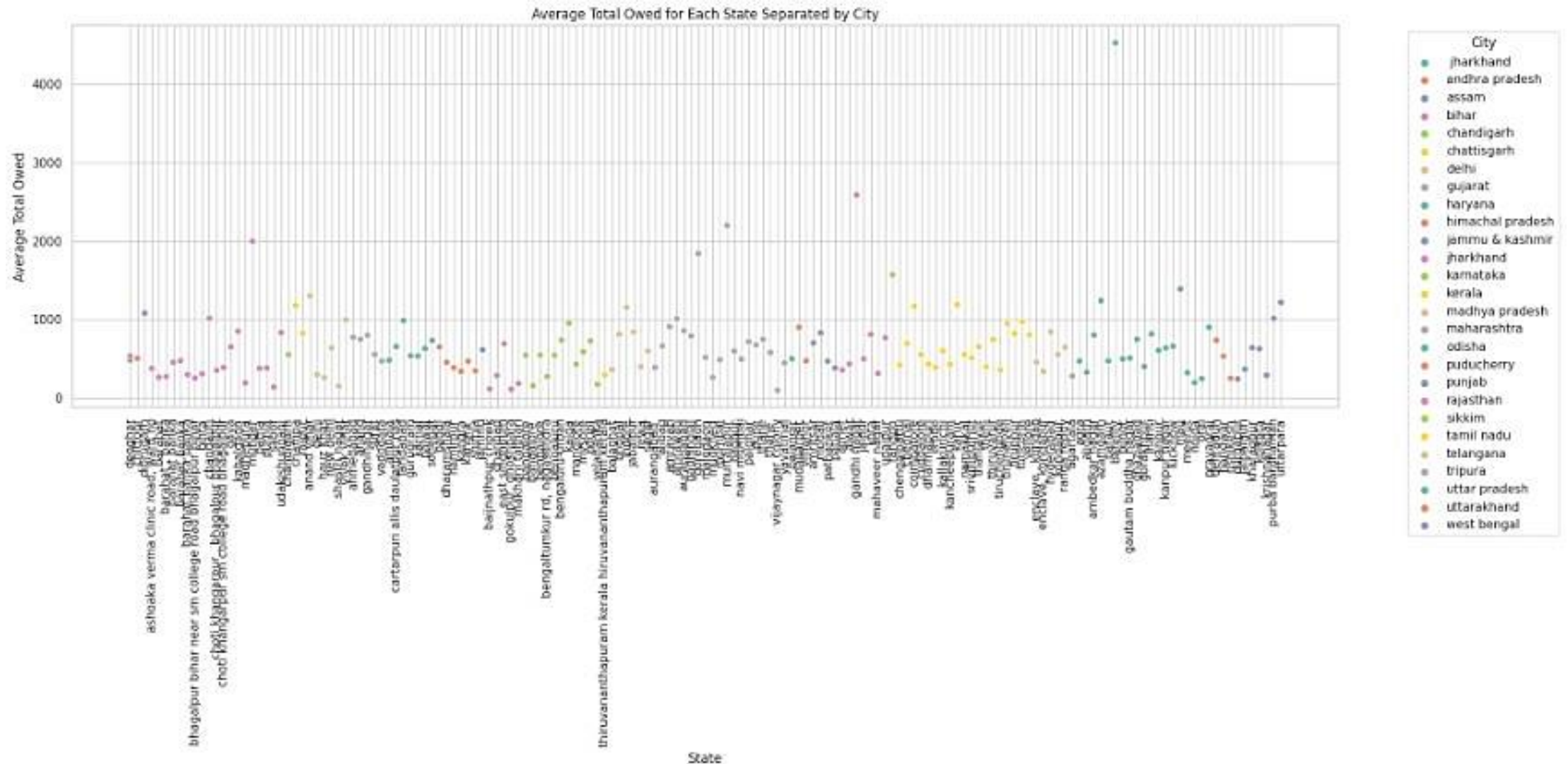
# Now check for differences in orders in different clusters
product_price_cluster_crosstab = pd.crosstab(df['Price Category'], customer_data['Cluster'])
chi2, p_value, _, _ = chi2_contingency(product_price_cluster_crosstab)
print(f"Chi-square result for product price: chi2 = {chi2}, p-value = {p_value}")
```

Chi-square result for product price: chi2 = 7.4051282051282055, p-value = 0.11596623719893992

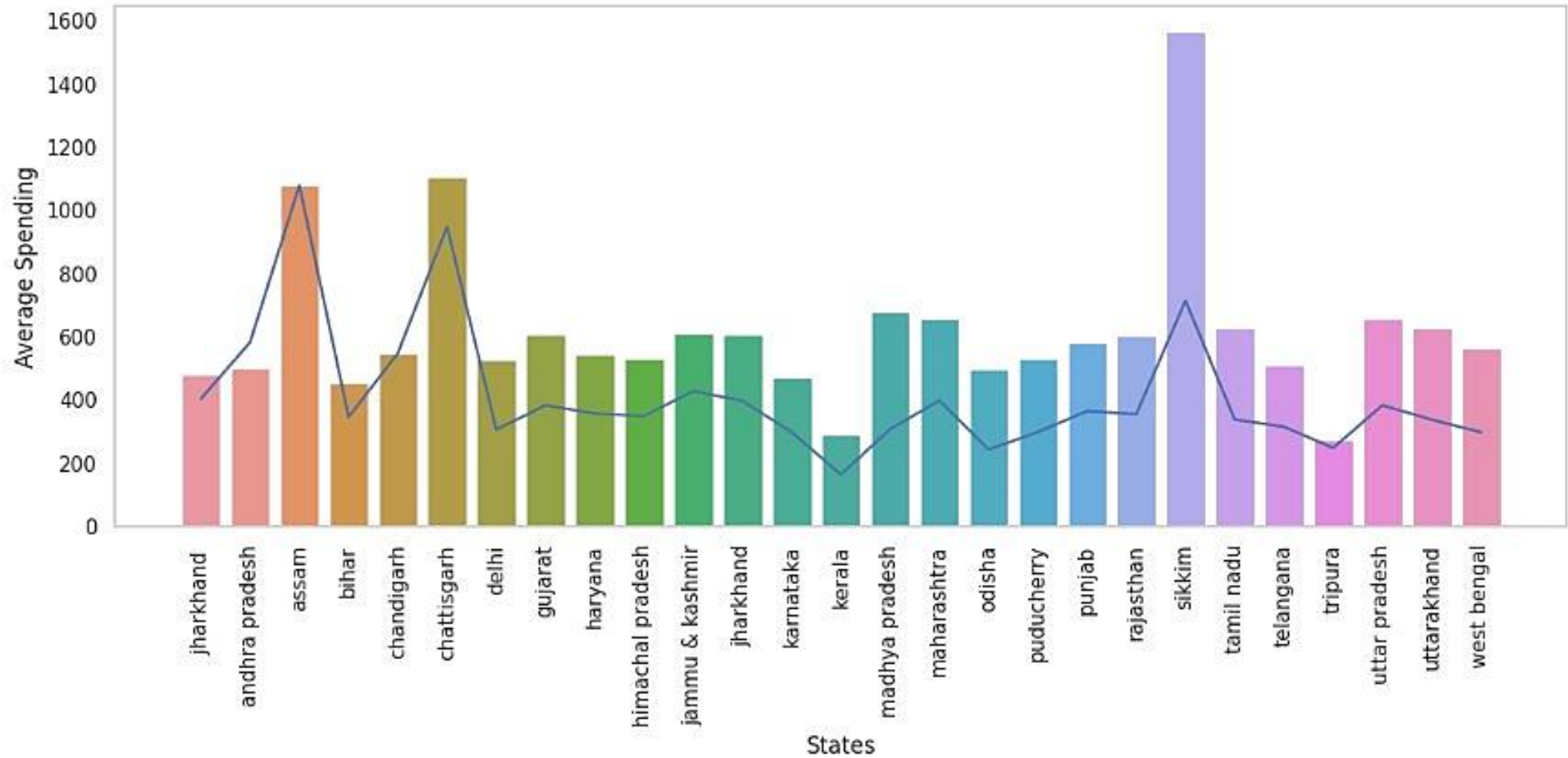
ORDER AND SPENDING ACROSS GENDER



AVERAGE TOTAL OWED ACROSS STATES



AVERAGE SPEND ACROSS STATE



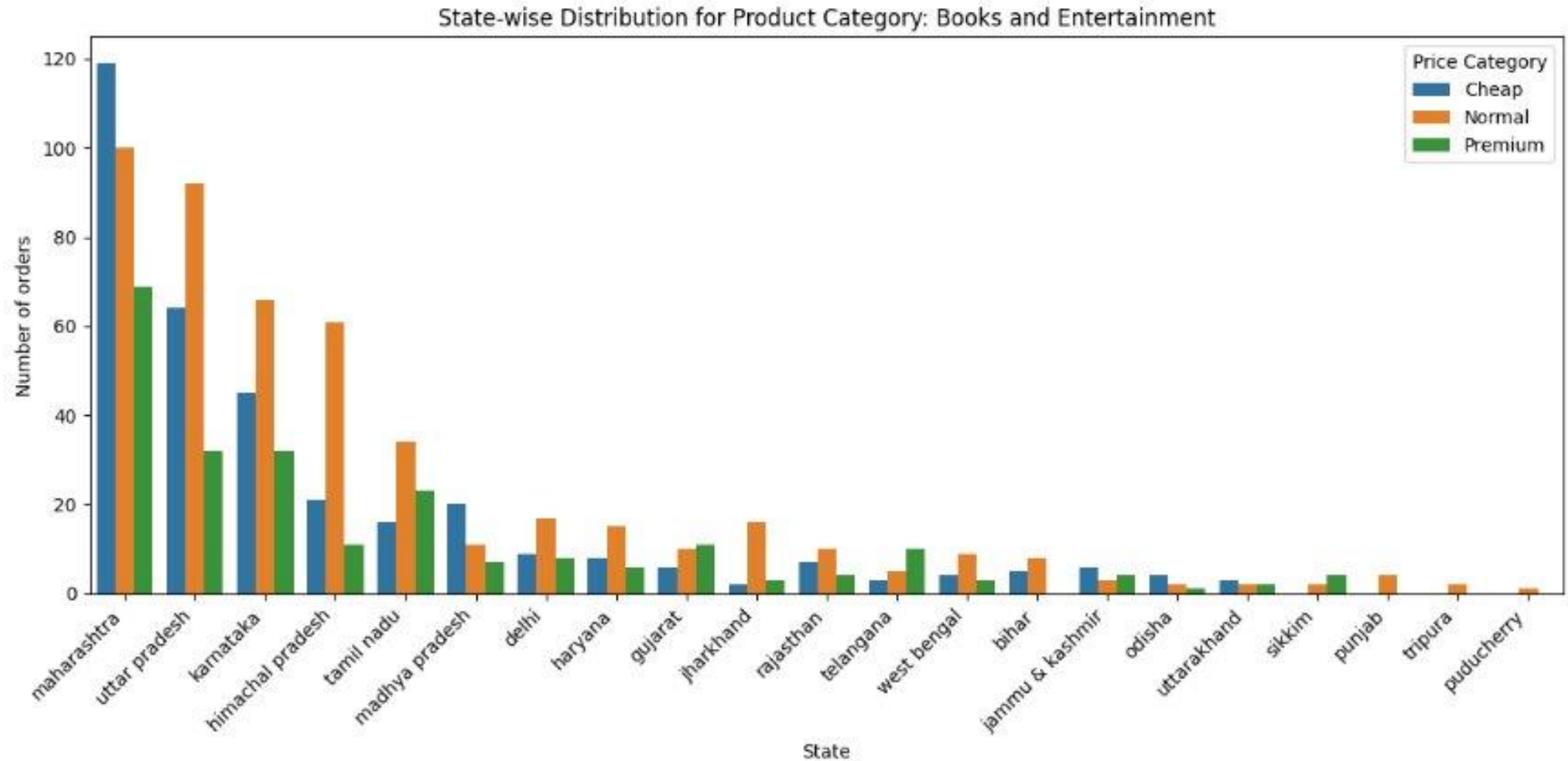
SPENDING CAPACITY ACROSS STATES

For each product category we have further sub categorized them based on

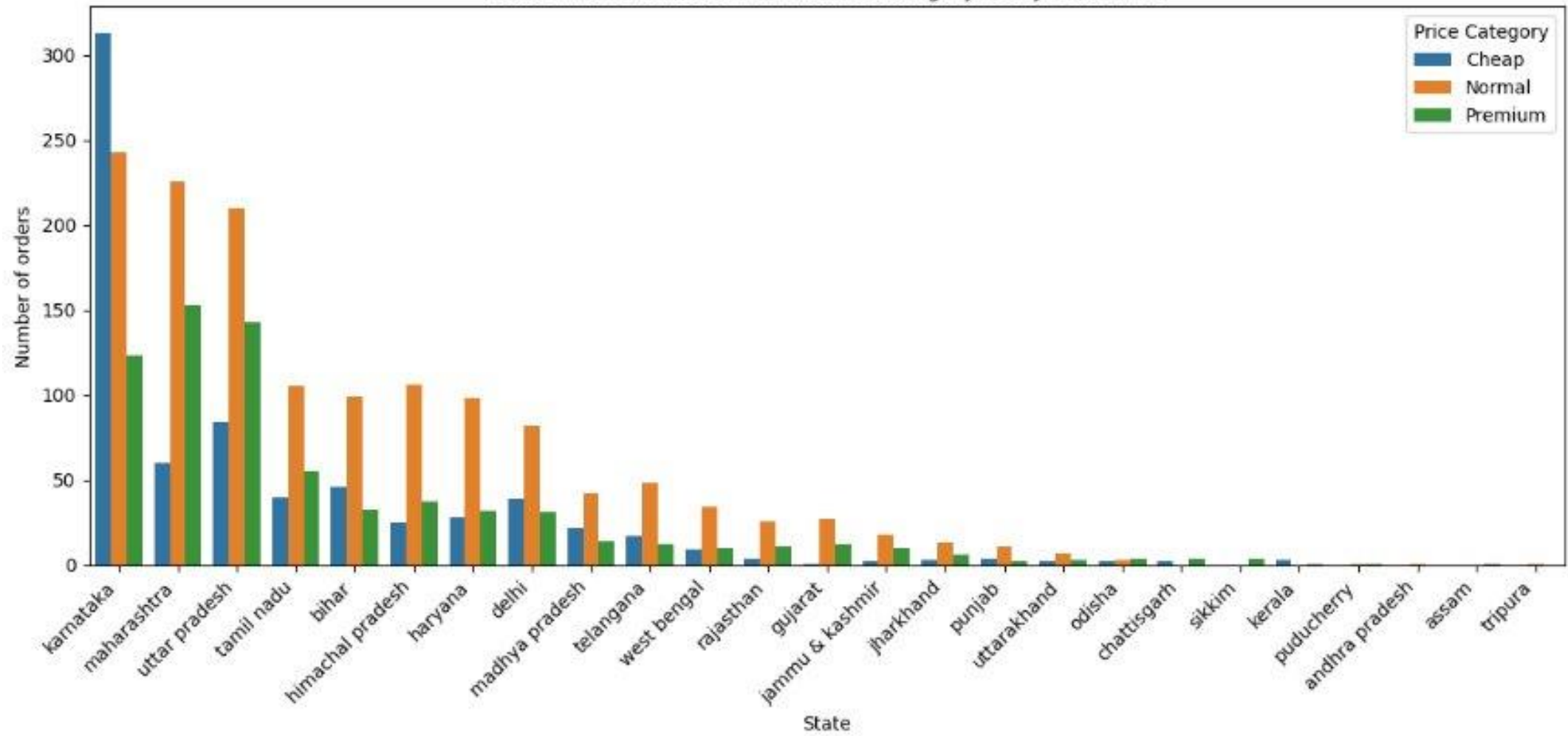
Unit price of a particular category :

1. **Cheap product** : If the price lies below 25th percentile
2. **Normal Product** : If the price lies between 25th and 75th percentile
3. **Premium Product** : If the price lies above 75th percentile

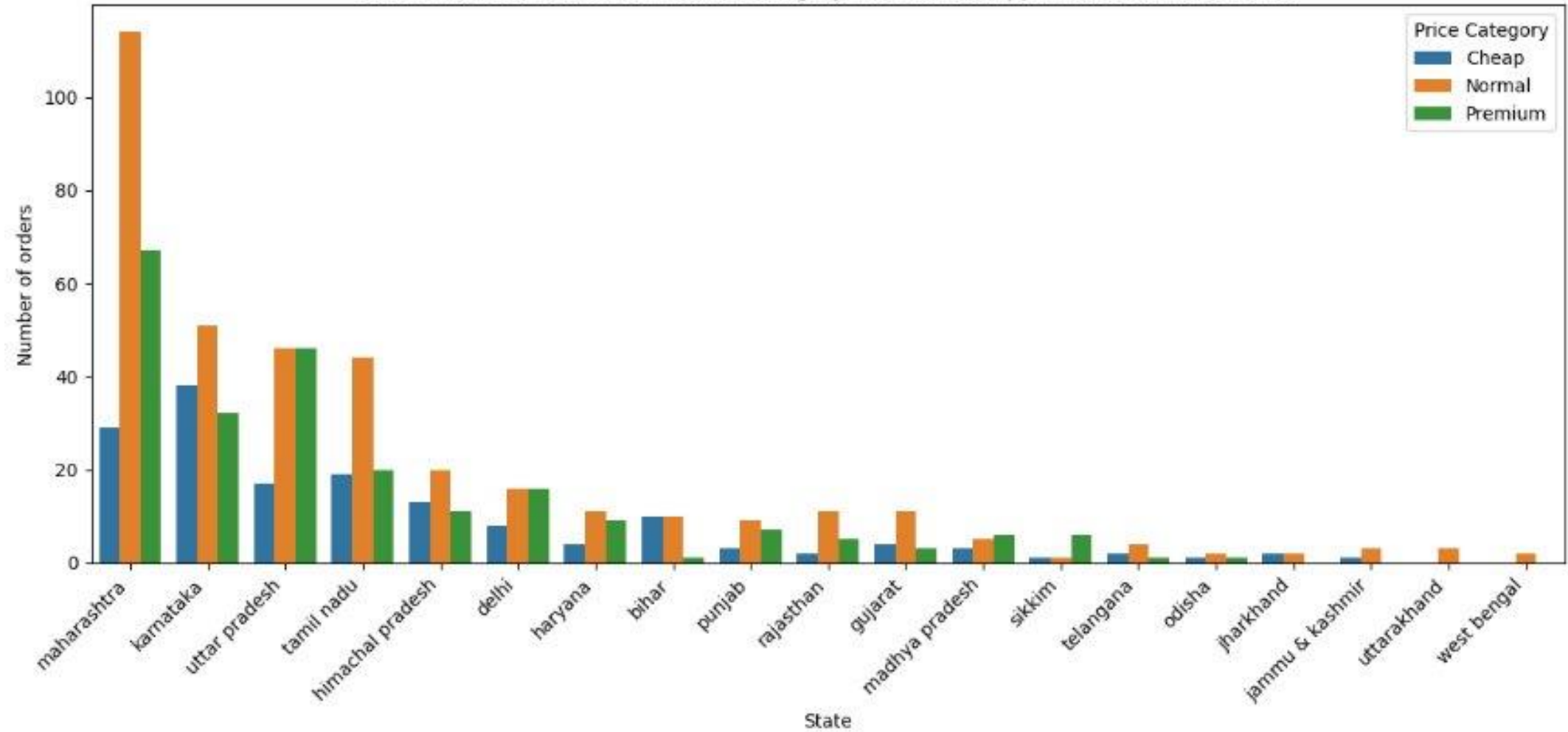
PRODUCT CATEGORY WISE PLOTS



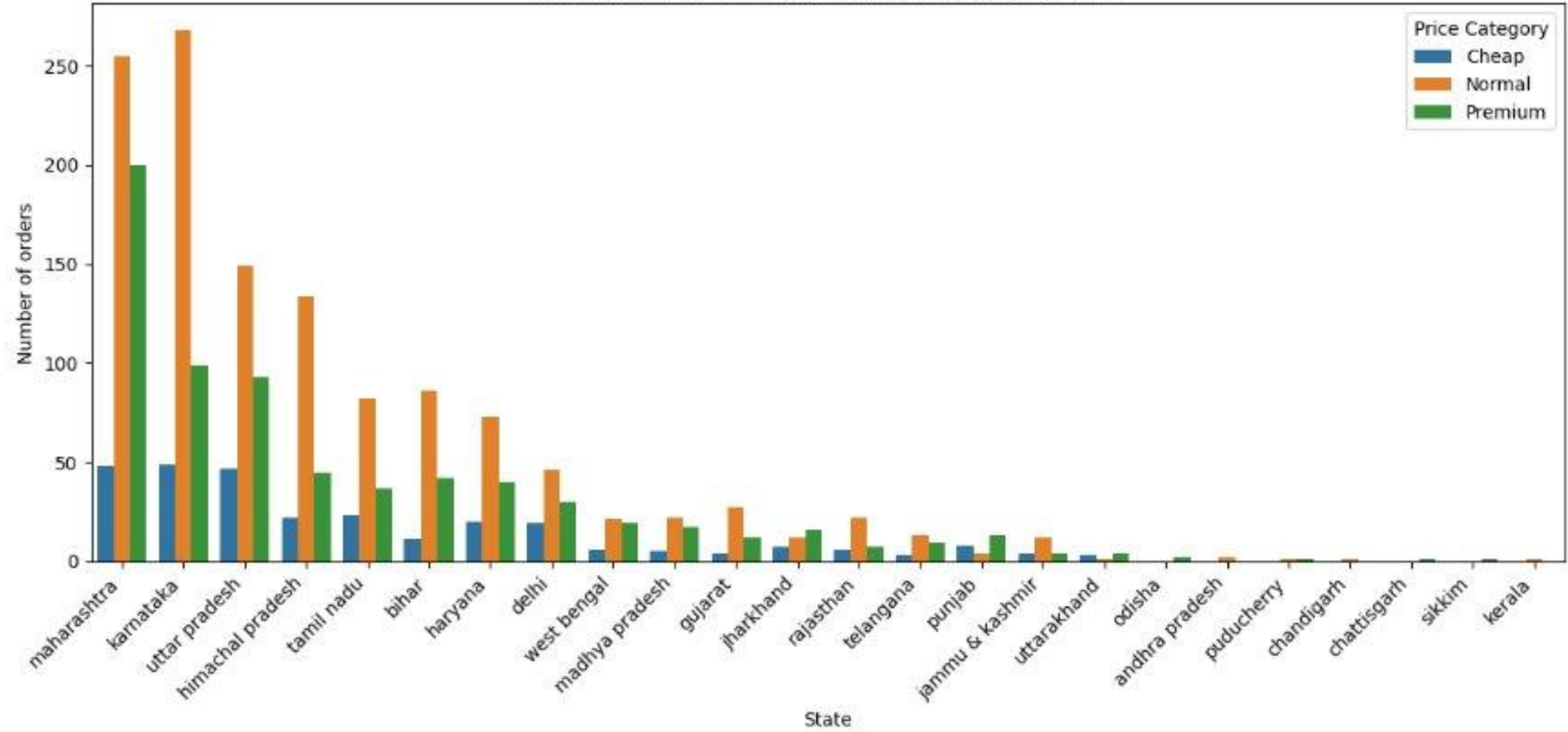
State-wise Distribution for Product Category: Daily Essentials



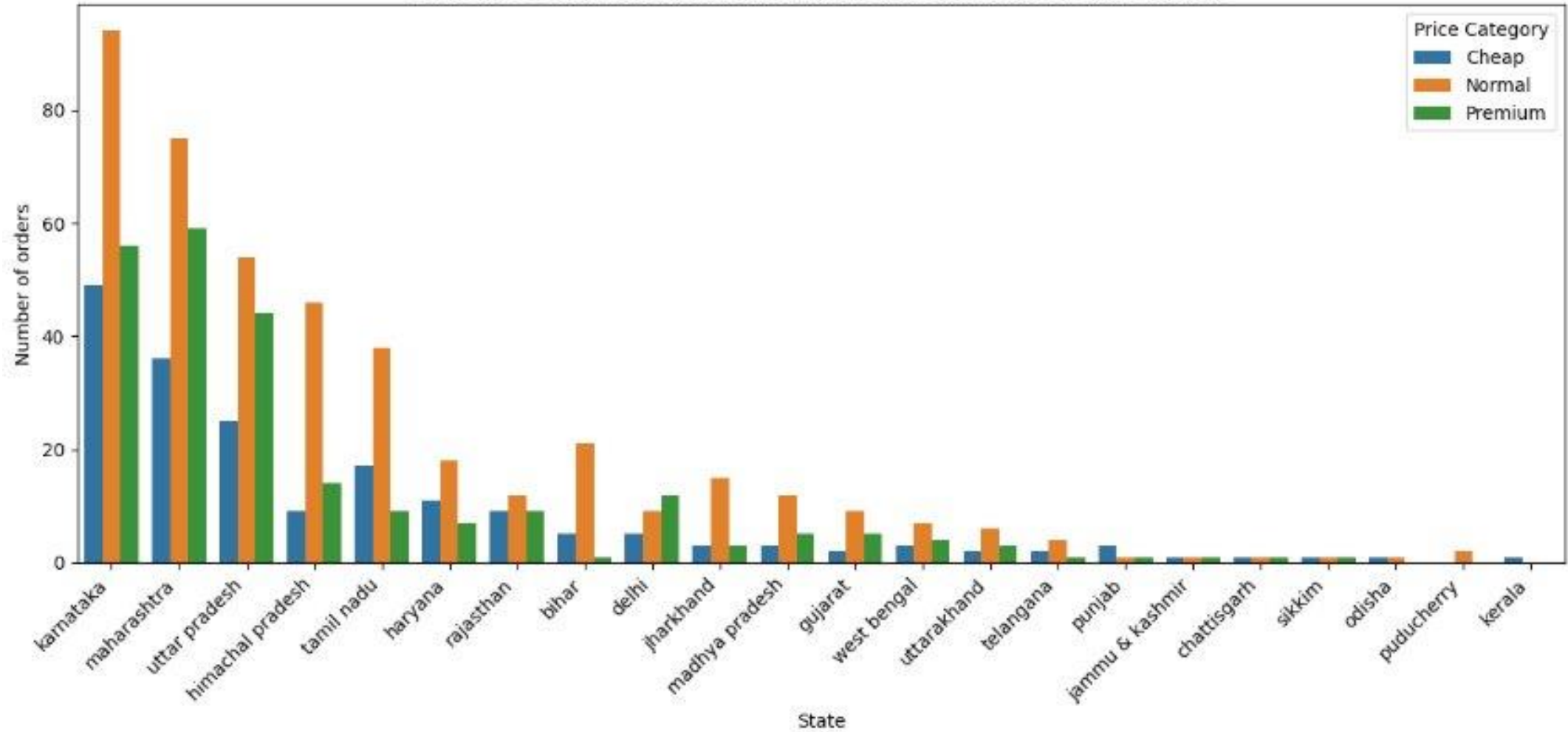
State-wise Distribution for Product Category: Electronics, Appliances and Accessories



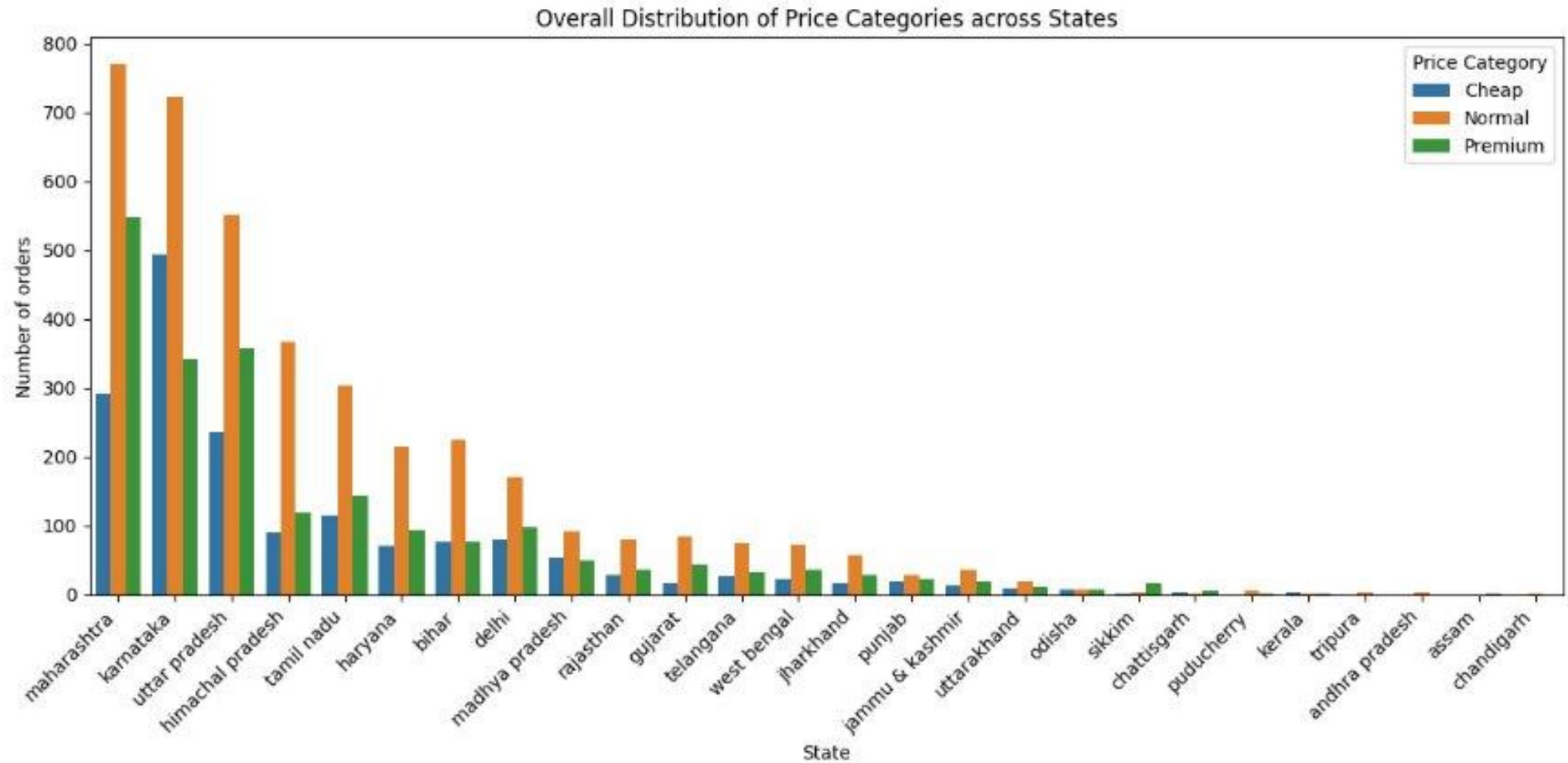
State-wise Distribution for Product Category: Fashion



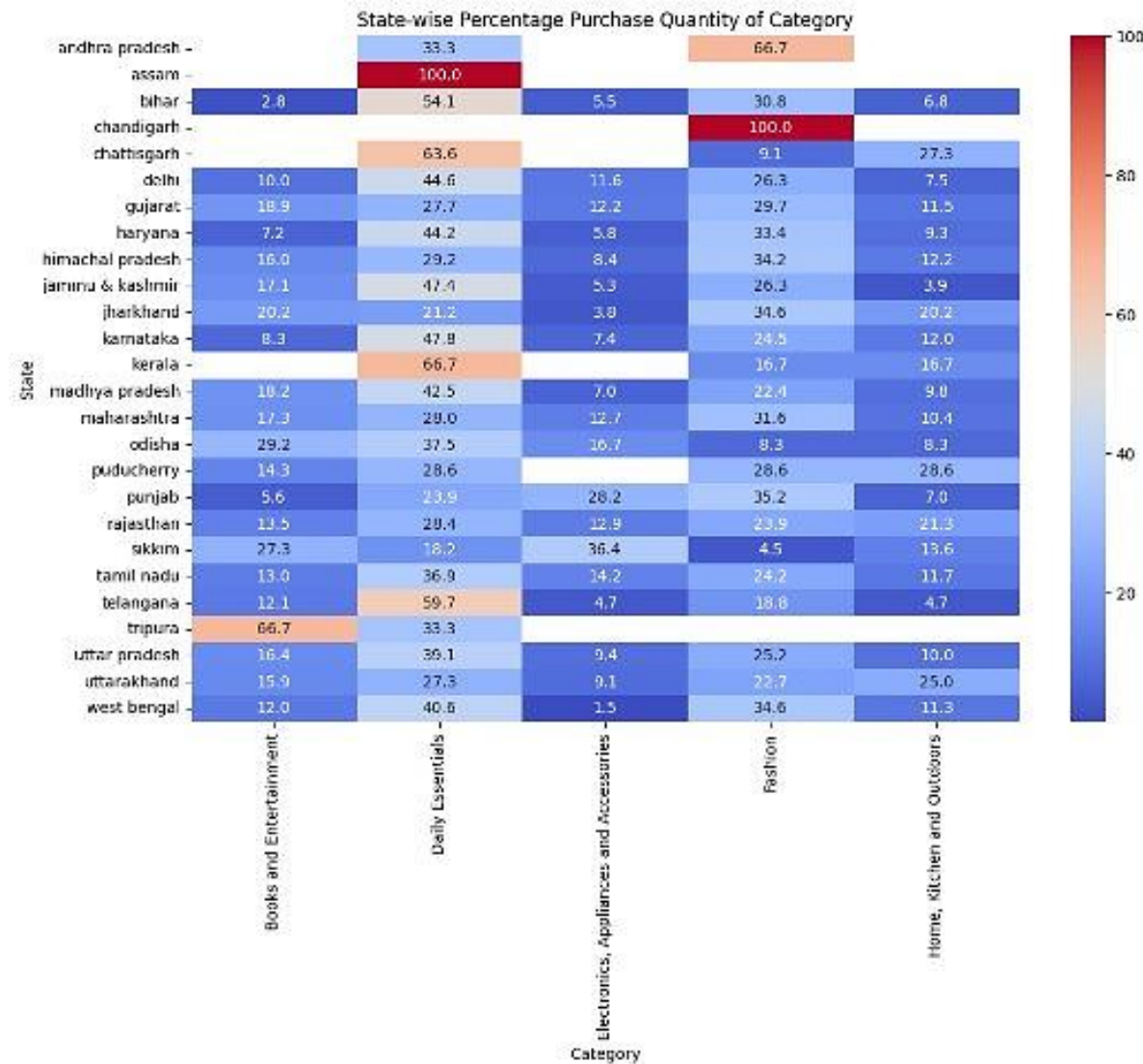
State-wise Distribution for Product Category: Home, Kitchen and Outdoors



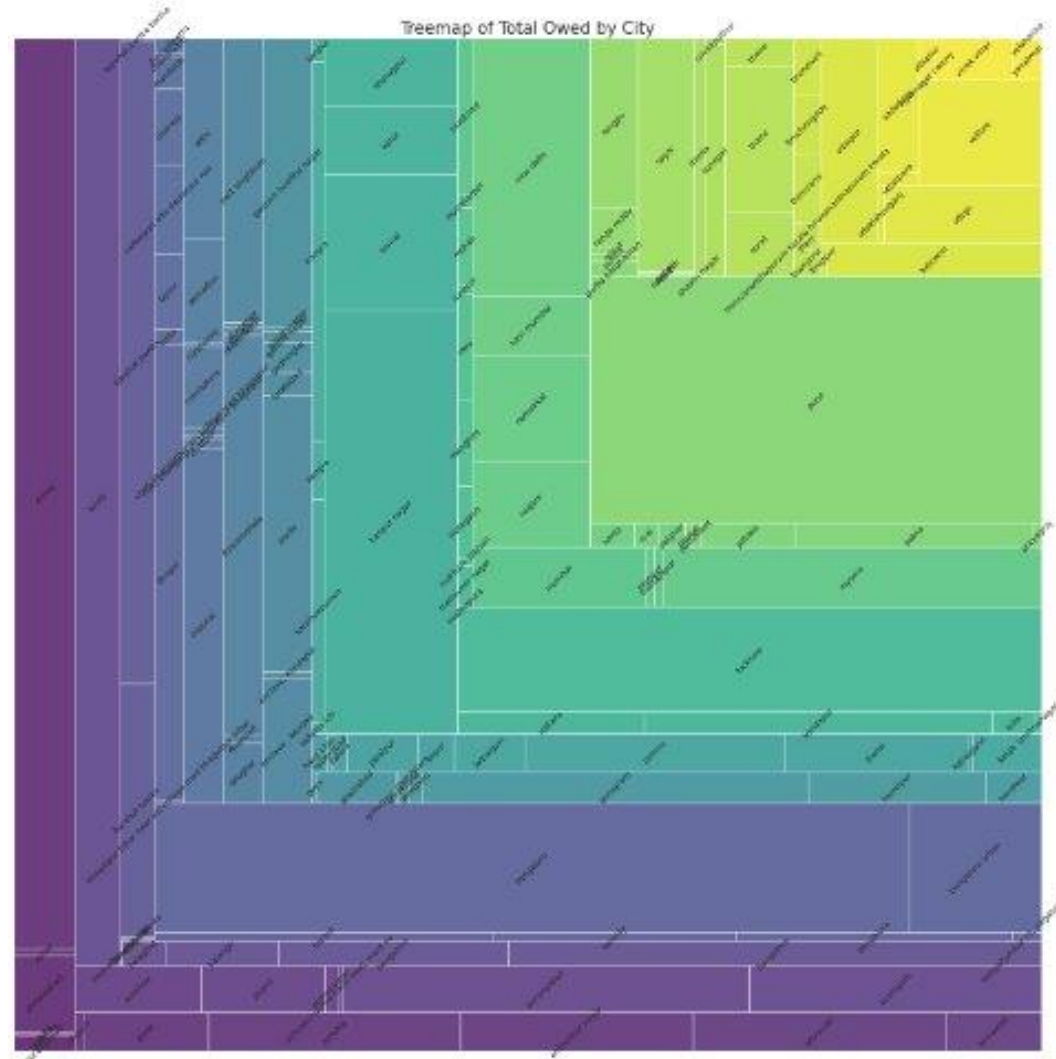
OVERALL DISTRIBUTION



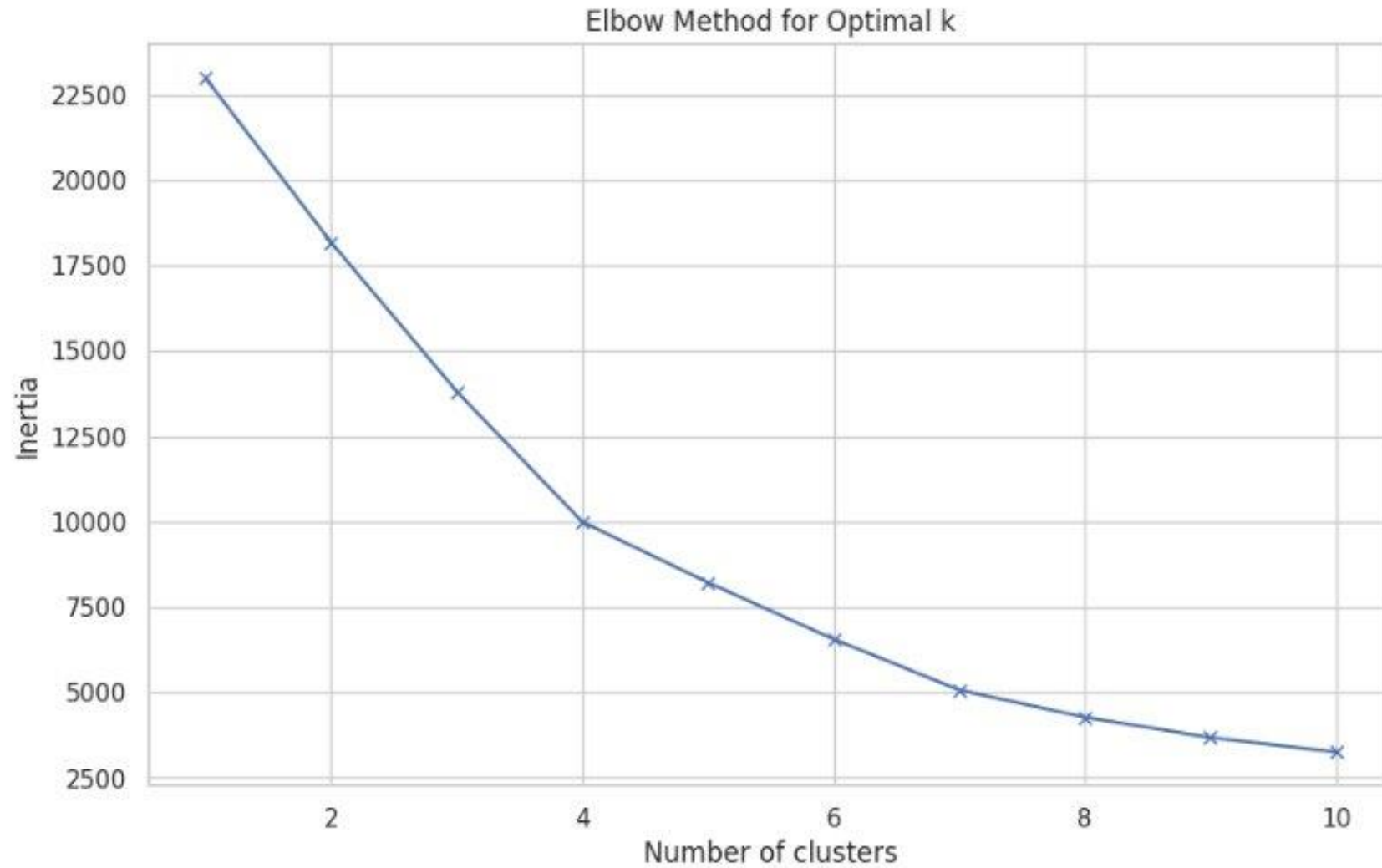
STATE WISE % DISTRIBUTION OF ORDER CATEGORY



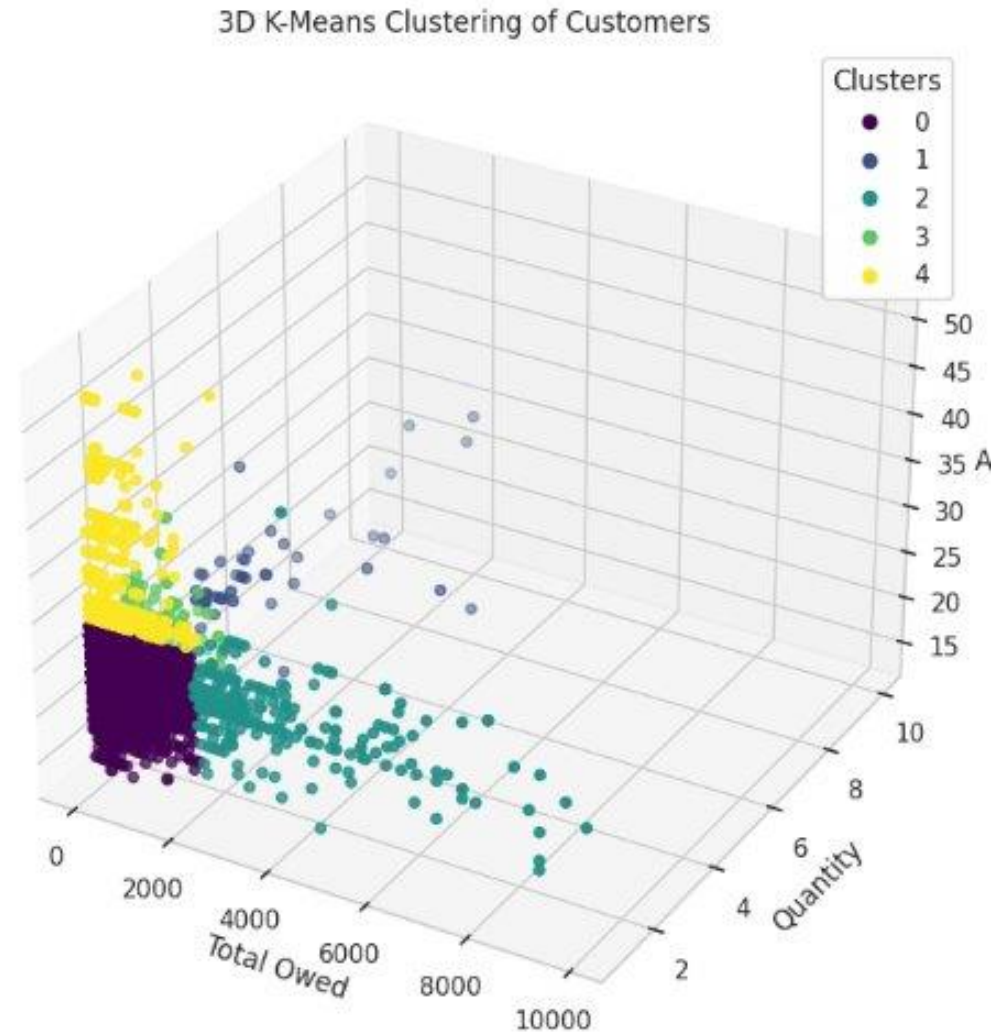
TREE MAP OF TOTAL OWED ACROSS CITY



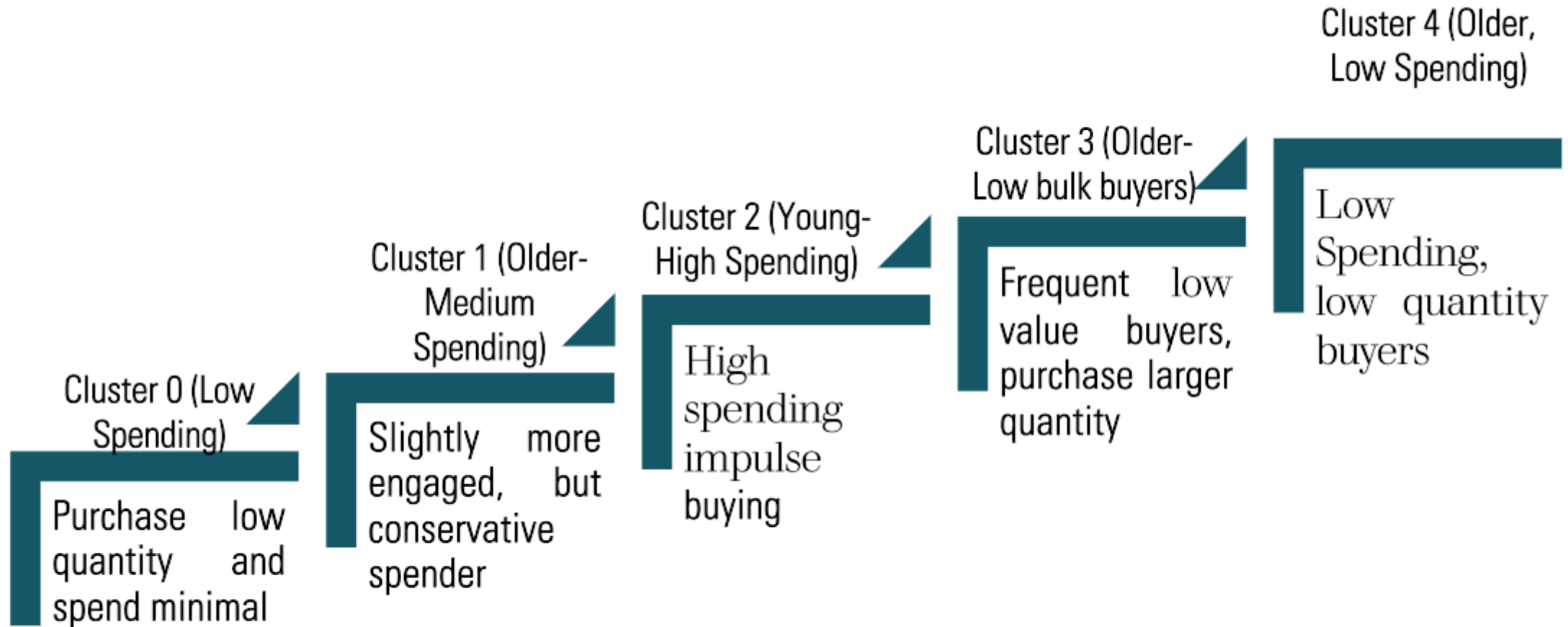
OPTIMAL K ESTIMATION (ELBOW METHOD)



3D PLOT OF K – MEANS CLUSTERING



SEGMENTATION BASED ON K-MEAN CLUSTERING



MARKETING STRATEGY

Cluster 0	Highlight cheap costs. Offer discounts/cashback
Cluster 1	Offer bundled deals like buy-1-get-1 free Appear on top of search bars
Cluster 2	Reinforce brand loyalty by giving exclusive discounts Target with trendy, fashion forward,
Cluster 3	VIP membership offering premium support, unique offers Retention by post purchase follow ups, better post sale service
Cluster 4	Dedicated account manager for building strong customers relationship Invite for participation in beta tests or provide feedback on new products

THANK
YOU